

H. String Mood Updates

time limit per test: 2 s
memory limit per test: 256 megabytes

Limak can be either happy or sad. His mood changes (or stays same) when he reads an English uppercase letter. Letters **S** and **D** always make him sad, **H** makes him happy, and every vowel (**A E I O U**) flips his mood. Other letters do nothing. Limak is happy right now and he wants to be happy after reading a string.

Let a *valid* character denote a question mark **?** or an uppercase English letter **A–Z**.

You are given a pattern-string s with n valid characters. If there are, say, x question marks, there are 26^x ways to replace them with uppercase letters **A–Z**. Your task is to count ways of replacing so that Limak would be happy after reading a new string (if he's happy before).

Additionally, there are q updates of form „ $i\ c$ “, each modifying one character of the pattern s . You need to print the answer before updates and after every update, modulo $10^9 + 7$.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 200\,000$) — the length of the pattern and the number of updates.

The second line contains the initial pattern s with n valid characters.

Each of the following q lines contains an integer i ($1 \leq i \leq n$) and a valid character c . The update changes the i -th character of s to c . No update tries to change a character into the same character.

Characters in s are 1-indexed. After you change a character, this change stays on (updates are not temporary).

Output

Print $q + 1$ lines, each with the current answer modulo 1000000007.

Example

input	Copy
2 5 A? 2 0 1 H 1 ? 2 ? 2 H	
output	Copy
6 1 0 7 403 26	

Note

The pattern-string s is initially **A?** and then after every next update: **AO**, **HO**, **?O**, **??**, **?H**.

For the initial pattern **A?**, there are six good ways to replace all question marks: **AA**, **AE**, **AH**, **AI**, **AO**, **AU**. The first letter is a vowel so it flips Limak's mood from happy to sad. The second letter must be a vowel or **H**.

Matrix Exponentiation

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
336330202	Aug/31/2025 07:29	Happy New Year!

